ASSESSMENT OF INDOOR AIR QUALITY IN AN EDUCATIONAL INSTITUTION {PVPSIT}

Submitted by

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Why We Have To Assess The IAQ ??

స్వచ్ఛమైన గాలి రూ.1499 మాత్రమే

వ్యాపారానికి ఏదీ అతీతం కాదు. ఢిల్లీలో గాలి కాలుష్యం తీవ్రం కాగా.. స్వచ్ఛ గాలి పేరుతో పలు కంపెనీలు వ్యాపారం చేస్తున్నాయి. ఓ ఆస్ట్రేలియా ఓ కంపెనీ 7.5 లీటర్ల గాలిని ₹1499కి విక్రయిస్తుండగా.. ఉత్తరాఖండ్‌కు చెందిన మరో కంపెనీ 160సార్లు పీల్చుకునే గాలిని ₹550కు అందిస్తోంది. దీన్ని కొంటే ఓ బాటిల్, మాస్క్ మాదిరిగా ఉండే పంపు వస్తుంది. దాన్ని ముఖానికి అమర్చి.. బటన్ నొక్కితే స్వచ్ఛమైన గాలి ముక్కుకు అందుతుంది.

CONTENT

- Abstract
- Objectives
- Literature Review
- Introduction To IAQ
- Methodology
- Parameters in Air
- Health Effects
- Results
- Survey on IAQ
- Ways to Improve IAQ
- Conclusion

ABSTARCT

- Air pollution sources in indoor area are one of the main factors for reducing indoor air quality of locations. In the current research, factors affecting indoor air quality and its symptoms are evaluate as case study in an Educational institution(PVPSIT) in Vijayawada.
- In order to conduct this descriptive-analytical study, gas pollutants affecting indoor air quality were determined using portable analysis systems. Indoor Air Quality Toolkit was used in order to study the Indoor air quality.
- Hence, we aimed to evaluate the indoor air quality of all class rooms, Laboratories and Faculty rooms(staff rooms) of all the departments in our institution.
- We prospectively measured indoor air quality using air quality device in different areas of our Institution. In this process, CO2, Temperature and Humidity will be measured.

OBJECTIVES

- TO Assess the Indoor Air Quality in an Educational Institution.
- To collect the Indoor Air Quality parameters such as CO2, Temperature and Humidity during the Rainy, Winter and Summer seasons.
- To analyze and evaluate the Indoor Air Quality parameters in three seasons.
- To compare the values of Indoor Air Quality parameters of classrooms, staffrooms and laboratories of each and every department individually in three seasons.
- To conduct a survey on Indoor Air Quality in PVPSIT.
- To suggest the measures to improve the poor Indoor Air Quality based on the analysis.

We will prospectively measure indoor air quality in number of locations in PVPSIT i.e.,

Class Rooms – 48

Staff Rooms – 68

Laboratories – 51

Grand Total – 167

LITERATURE REVIEW

1.Indoor air quality in educational institutions in Beijing: Field tests, problems and recommendations

Authors: Chaorui Cai, Zhiwei Sun and Louise .

This study investigated indoor CO2 and PM2.5 real-time concentrations together with air temperature, humidity and air-cleaning effectiveness of 33 classrooms in 21 educational institution in Beijing, China by natural ventilation with air cleaners (NV system) compared to mechanical ventilation with fresh air ventilators (MV system).

2. Indoor air quality and health in Institutional building: A critical review for developing the roadmap for the future school environment

Authors: Sasan Sadrizadeh, Runming Yao, Feng Yuan and Hazim Awbi

In this article, they analyzed indoor air quality and health in schools, with the primary focus on ventilation, thermal comfort, productivity, and exposure risk. This study conducts a comprehensive review to summarizes the existing knowledge to highlight the latest research and solutions and proposes a roadmap for the future school environment.

3. Overheating and indoor air quality in primary schools in the UK

Authors: Sara Mohamed, Lucelia Rodrigues, Siddig Omer and John Calautit

In this work, the authors conducted field studies to assess the indoor air quality and overheating levels in eight recently built classrooms in two new low-carbon primary schools during heating and non-heating seasons. Monitored variables included globe temperatures, air temperatures, surface temperatures, carbon dioxide (CO2), and particles matter with a diameter of less than 2.5 mm (PM2.5), airstream velocity and relative humidity.

4. Indoor air quality in the primary school of China—results from CIEHS 2018 study

Authors: Yuan-duo Zhu, Xu Li, Lin Fan, Li Li, Jiao Wang, Wen-jing Yang, Lin Wang, Xiao-yuan Yao, and Xian-liang Wang .

Their results suggest that indoor air pollution in classrooms was a severe problem in Chinese primary schools. It is necessary to strengthen ventilation in the classroom to improve indoor air quality. What’s more, a healthy learning environment should be created for primary school students. The ratios of indoor to outdoor (I/O) of PM2.5 were calculated in each classroom to identify whether the indoor environment the pollutants comes from outdoors

5. Quantifying indoor air quality determinants in urban and rural nursery and primary schools.

Authors: P.T.B.S. Branco , M.C.M. Alvim-Ferraz , F.G. Martins and S.I.V. Sousa.

This study mainly 16 aimed to quantify determinants of selected IAP, in nursery and primary schools from both 17 urban and rural sites, accounting for seasonal variations. In 101 indoor microenvironments 18 (classrooms, bedrooms and canteens) from 25 nursery and primary schools, CO2, CO, HCOH, 19 NO2, O3, total volatile organic compounds, PM1, PM2.5, PM10, total suspended particles

6. Assessment of indoor and outdoor air quality in primary schools of Cyprus during the COVID–19 pandemic measures in May–July 2021.

Authors: Corina Konstantinou, Andria Constantinou and Eleni G. Kleovoulou

The study took place during May–July 2021 when a schoolspecific COVID-19 protocol was in place. Questionnaire-based characteristics of schools/classrooms were collected along with 24/48-h long IAQ monitoring of air temperature, relative humidity (RH), particulate matter (PM), carbon dioxide (CO2) and volatile organic compounds (VOCs), using low-cost sensors. Mixed effect models assessed the IAQ determinants during school hours. Indoor PM, temperature, RH and VOCs increased with progressing school periods in the day, while indoor CO2 decreased.

7. Investigation of indoor and outdoor air quality in a university campus during COVID-19 lock down period

Authors: Mansoor Ahmad Bhat, Fatma Nur Eraslan, Alaa Awad and Semra Malkoc

This study intends to give a glimpse to quantify the impact of a pandemic on air quality in different indoor environments in a university campus in Eskis¸ehir, Turkey and calls for follow-up studies. Indoor concentrations were evaluated together with outdoor concentrations. In general, it can be said that the calculated I/O ratios for SO2, NO2, O3, bacteria, and fungi were less than 1 in most indoor environments.

8. A comprehensive analysis of the intervention of a fresh air ventilation system on indoor air quality in classrooms

Authors: Yu-HaoChen, Yen-PeiTu and Shih-YiSung

This study investigated CO2 and PM2.5 concentrations in 86 classrooms installed with air conditioners in elementary and middle schools in southern Taiwan, each additionally installed with a fresh air ventilation system equipped with an E11-class filter.

INTRODUCTION

- Air pollution is currently recognized as the single biggest environmental threat to human health.
- People spend 90% of their time in indoor environments, therefore maintaining adequate indoor air quality is essential to minimize negative health impacts. Institutions are no different, as Students and Faculty members spend long hours in the facilities subject to their inherent air quality.
- Indoor air quality (IAQ) is a complex and dynamic issue that is affected by different factors:

-Outdoor air Quality

-Indoor occupant density

-Indoor intrinsic emissions

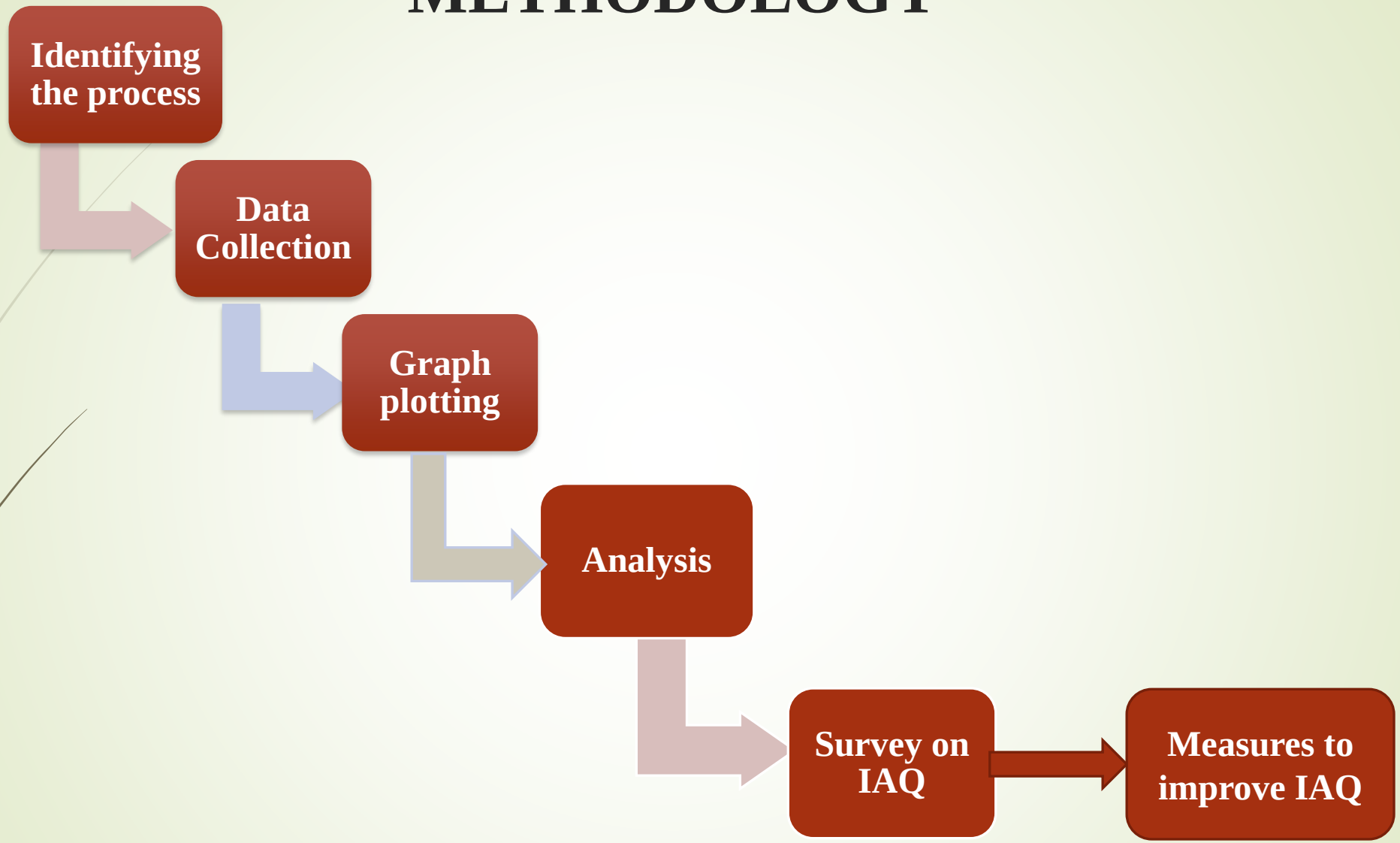
(e.g., equipment/furniture/coatings)

-Indoor activities

-Ventilation practices

- The effect of the indoor environment on people's lives, the performance of work and the public's health is of more concern than the outdoor environment because people spend about 70% to 90% of their time in rooms.
- It is well accepted that indoor environments air pollutants have adverse health effects on humans.
- Statistics through studies show that people spend most of their time in an indoor environment, with 87% of the time inside a closed building and 6% in a closed vehicle.
- As a result, people can be at risk with prolonged exposure to indoor air quality (IAQ).
- Besides, factors such as inadequate ventilation, growth of microorganisms on walls, in carpets, or the components of heating, ventilation and air-conditioning systems cause poor indoor air quality and might even threaten human health. This was the motivation behind the number of studies that have been conducted on the indoor environment quality (IEQ) in Institutions.

METHODOLOGY



LOCATIONS IN PVPSIT

Department	Class Rooms	Staff Rooms	Laboratories
CSE	6	12	7
IT	6	9	5
ECE	6	11	8
EEE	6	9	7
CE	3	8	7
ME	6	4	9
FED	12	11	8
MBA	3	4	-
TOTAL	48	68	51

Grand Total = 48+68+51 = 167

PARAMETERS IN AIR

- Carbon dioxide
- Temperature
- Humidity
- Nitrogen dioxide
- Sulphur dioxide
- Carbon monoxide
- Lead
- Ozone
- Formaldehyde
- Methane
- Hydrogen
- Ammonia
- Hydrogen Sulphide
- Oxygen
- Argon
- Neon
- Xenon
- Nitrous oxide.

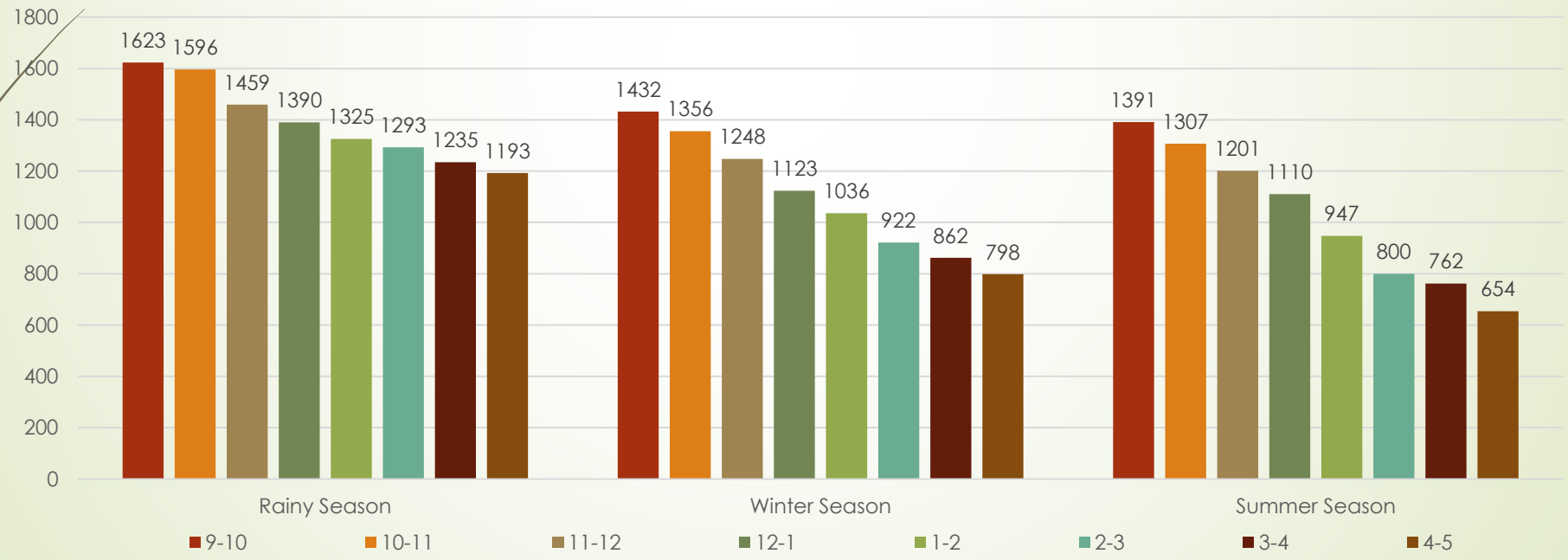
HEALTH EFFECTS

- Difficulty in Concentrating
- Illness, Allergies and Stress
- Cold
- Irritation of Eyes, Nose and Throat
- Headaches
- Rashes, Muscle pain and Fatigue
- Asthma
- Respiratory diseases, Heart diseases and Cancer
- Nasal congestion

RESULTS

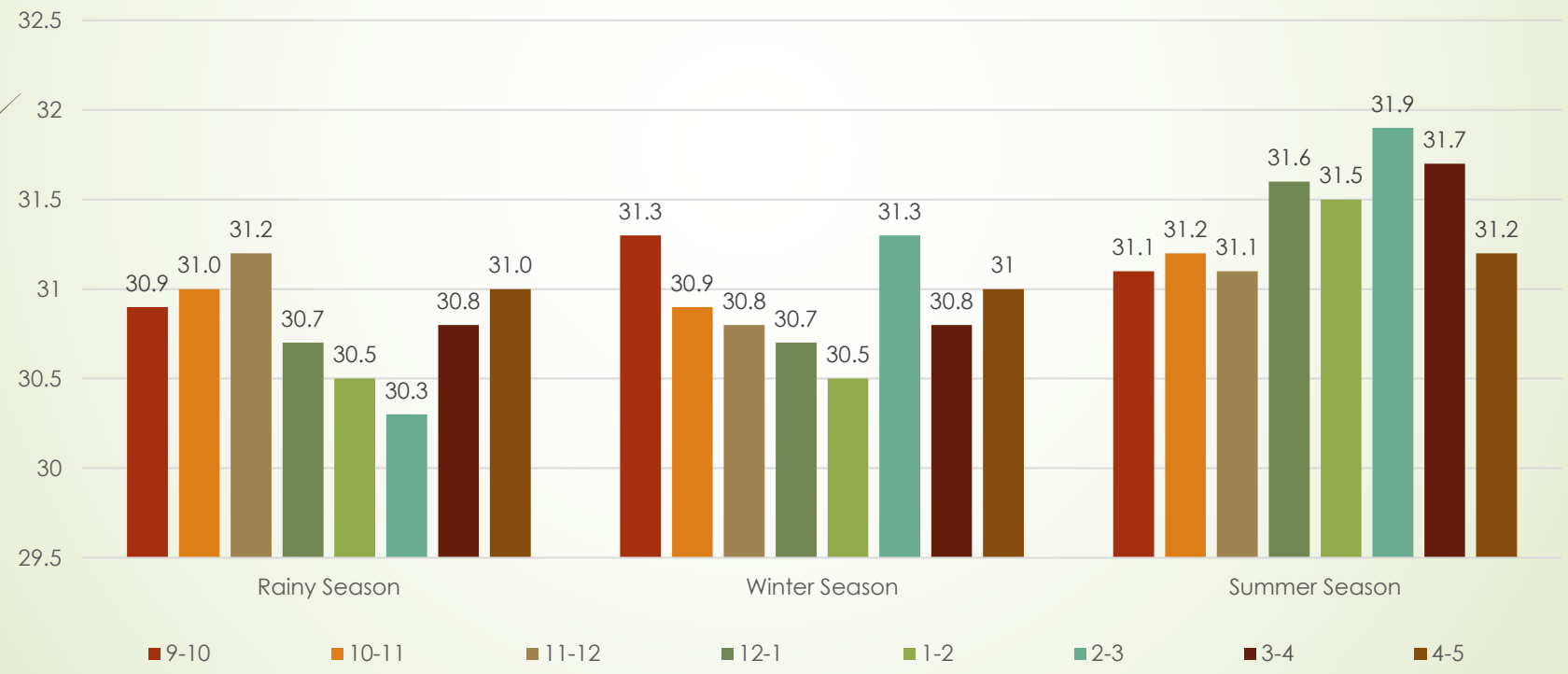
CO2 values in Final Year Class Room of CE Dept

Name of the Season	9-10	10-11	11-12	12-1	1-2	2-3	3-4	4-5
Rainy Season	1623	1596	1459	1390	1325	1293	1235	1193
Winter Season	1432	1356	1248	1123	1036	922	862	798
Summer Season	1391	1307	1201	1110	947	800	762	654



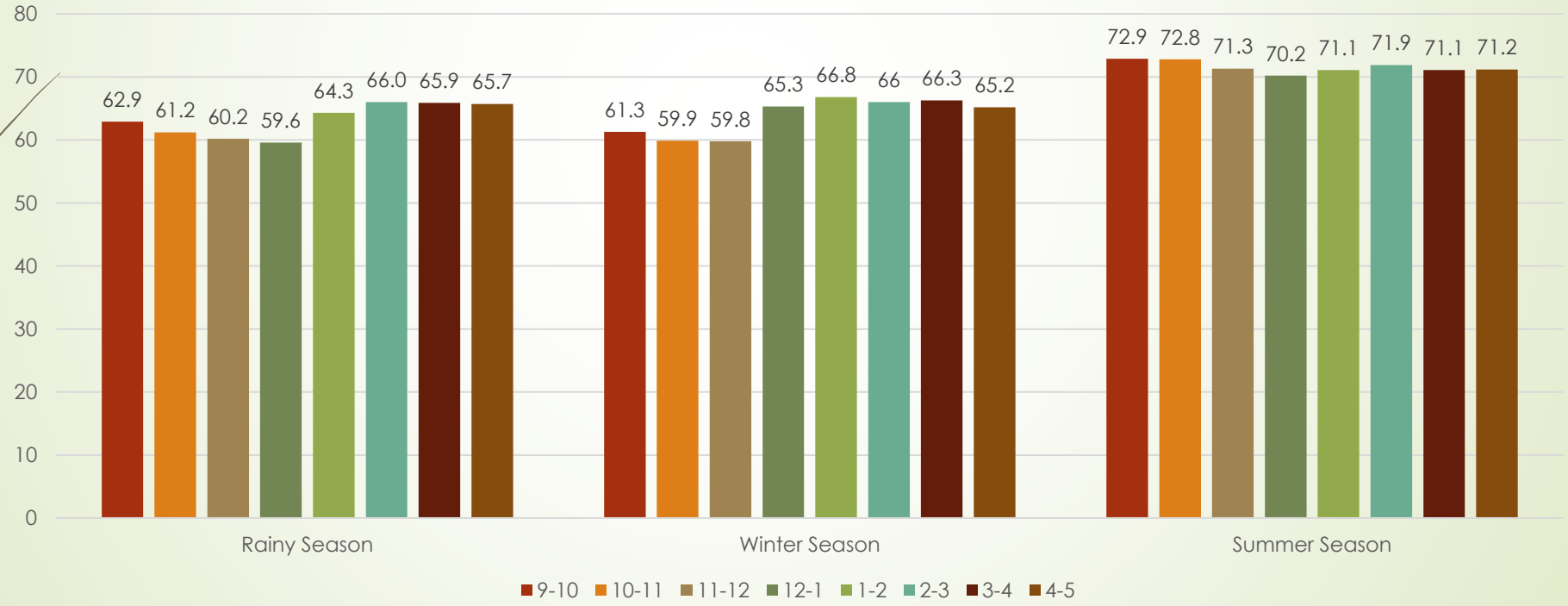
Temperature values in Final Year Class Room of CE Dept

Name Of The Season	9-10	10-11	11-12	12-1	1-2	2-3	3-4	4-5
Rainy Season	30.9	31.0	31.2	30.7	30.5	30.3	30.8	31.0
Winter Season	31.3	30.9	30.8	30.7	30.5	31.3	30.8	31
Summer Season	31.1	31.2	31.1	31.6	31.5	31.9	31.7	31.2



Humidity values in Final Year Class Room of CE Dept

Name Of The Season	9-10	10-11	11-12	12-1	1-2	2-3	3-4	4-5
Rainy Season	62.9	61.2	60.2	59.6	64.3	66.0	65.9	65.7
Winter Season	61.3	59.9	59.8	65.3	66.8	66	66.3	65.2
Summer Season	72.9	72.8	71.3	70.2	71.1	71.9	71.1	71.2



SURVEY ON INDOOR AIR QUALITY

Name of the Student *

Your answer

Age *

Your answer

Gender *

- Male
- Female
- Others

Register Number *

Your answer

Gmail Id *

Your answer

Room Number *

Your answer

Floor *

- Ground
- First
- Second
- Third

1. How do you find the classroom now? *

- Hot
- Cold

2. In general, during winter how is the classroom ? *

- Warm
- Cold

3. How is your state of health? *

- Bad
- Good
- Excellent

4. Last two weeks, do you have any respiratory illness ? *

- No
- Common Cold

5. Did the poor indoor air quality bother you during last hour? *

- Yes
- No

6. What bothered? *

- Dryness of air
- Humidity of air
- Too cold
- Too warm
- Draughtiness
- Stuffy air
- Too dusty
- Odour of the mold
- Fragrance/perfume

7. If you have any respiratory illness, does it decrease when you are not at college? *

- Yes
- No

8. Are there any signs of mold in the room? *

- Yes
- No

9. Are there any signs of humidity in the room? *

- Yes
- No

10. Has condensation been observed on window interiors? *

- Yes
- No

11. Air quality *

- Fresh
- Stuffy
- Odorless
- Smelly
- Clean
- Dusty

12. Satisfaction with indoor air quality *

- Dissatisfies
- Slightly Disatisfies
- Neutral
- Slightly Satisfies
- Satisfies

13. Impact of indoor air quality on work productivity *

- ◯ Interfered
- ◯ Slightly interfered
- ◯ Neither interfered/enhanced
- ◯ Slightly enhanced
- ◯ Enhanced

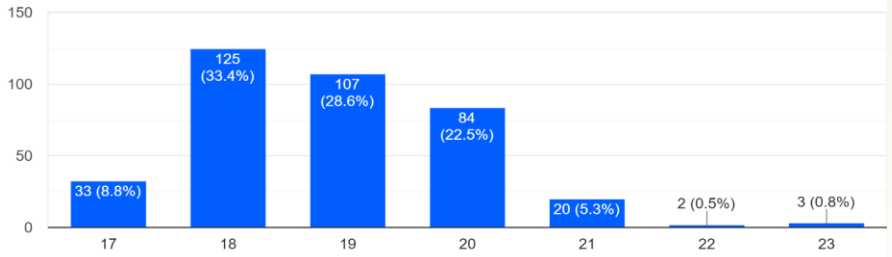
14. Self-reported health impacts *

- ◯ Never
- ◯ Sometimes
- ◯ Often

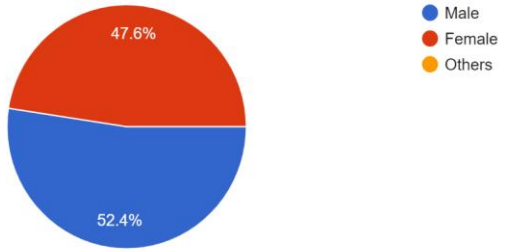
15. Would you have preferred that the windows were opened? *

- ◯ More often
- ◯ Less often
- ◯ It was good this way

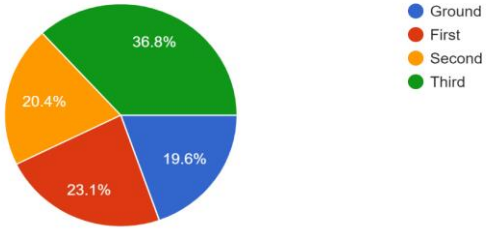
Age
374 responses



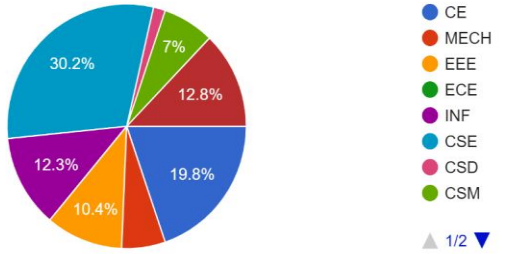
Gender
374 responses



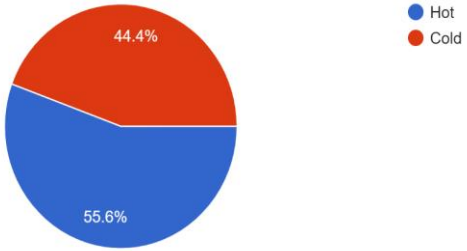
Floor
372 responses



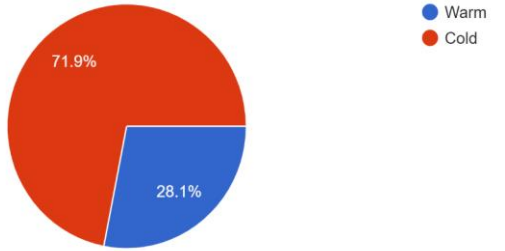
Department
374 responses



1. How do you find the classroom now?
374 responses

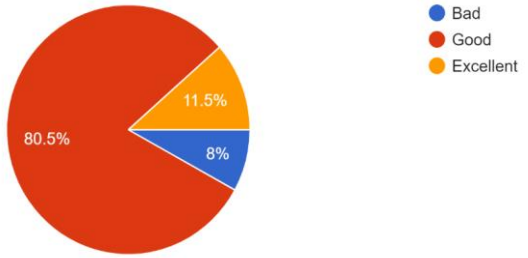


2. In general, during winter how is the classroom ?
374 responses



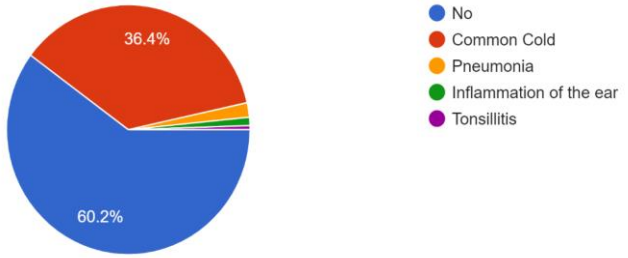
3. How is your state of health?

374 responses



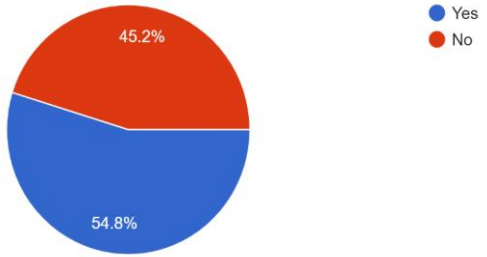
4. Last two weeks, do you have any respiratory illness ?

374 responses



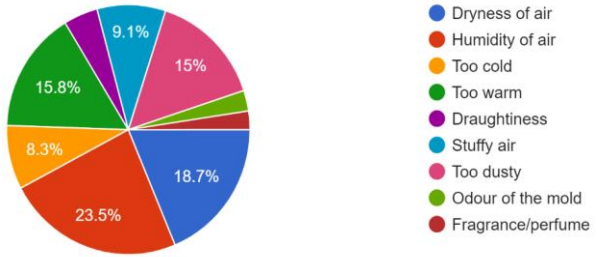
5. Did the poor indoor air quality bother you during last hour?

374 responses



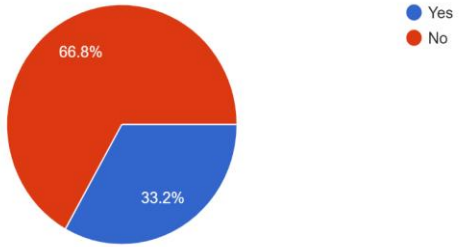
6. What bothered?

374 responses



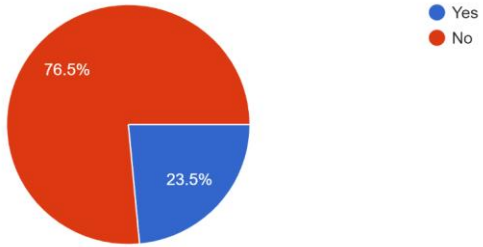
7. If you have any respiratory illness, does it decrease when you are not at college?

374 responses



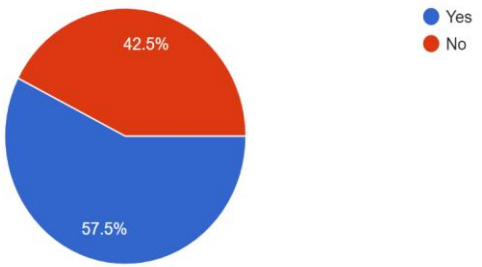
8. Are there any signs of mold in the room?

374 responses



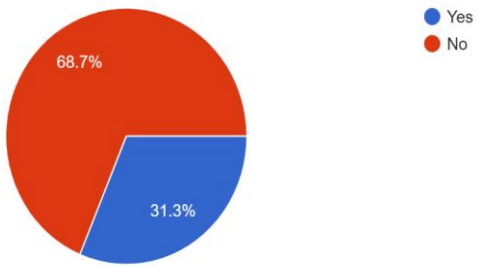
9. Are there any signs of humidity in the room?

374 responses



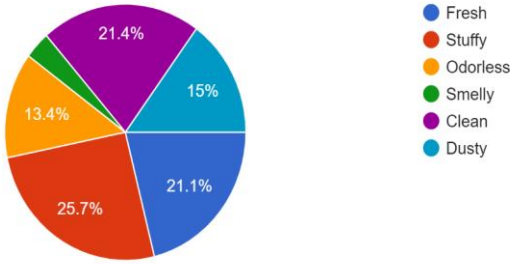
10. Has condensation been observed on window interiors?

374 responses



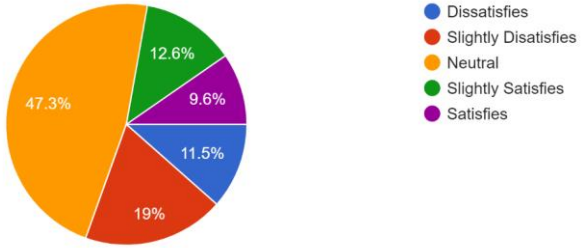
11. Air quality

374 responses



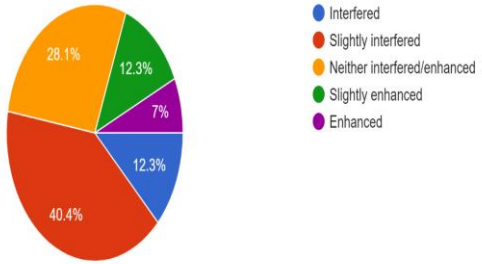
12. Satisfaction with indoor air quality

374 responses



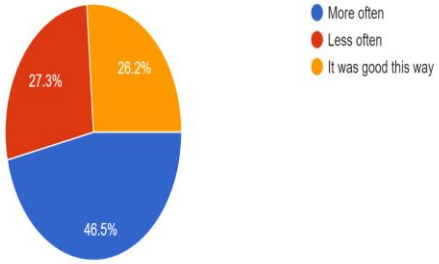
13. Impact of indoor air quality on work productivity

374 responses



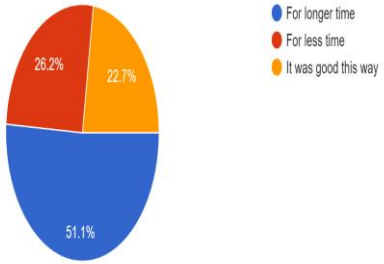
15. Would you have preferred that the windows were opened?

374 responses

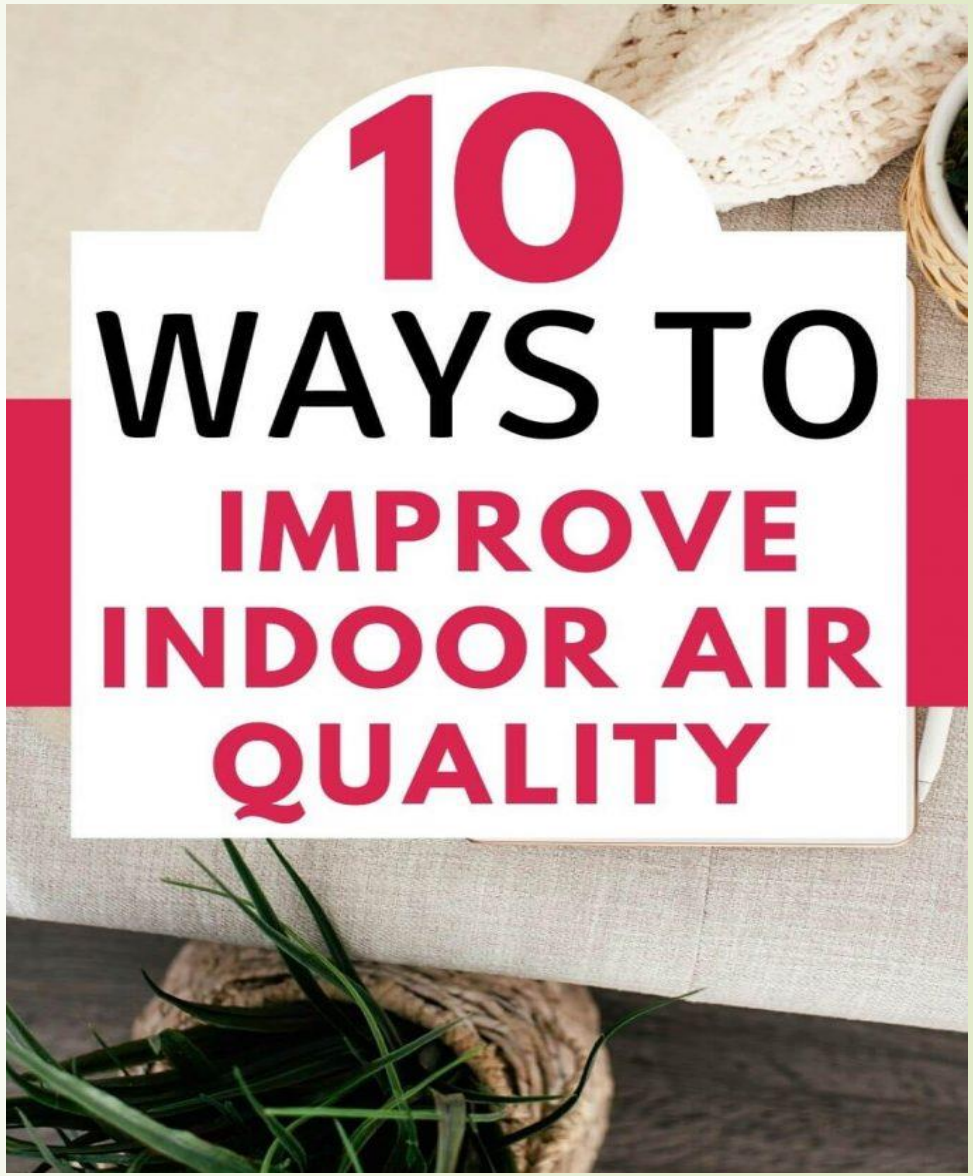


16. would you have preferred that the windows were kept opened ?

374 responses



1. By placing mini plants
2. Air Purifiers
3. Don’t cover up odors
4. Vacuum often
5. Use exhaust fans
6. Ban smoking
7. Open windows
8. Kept ventilators open
9. Try to kept the room dry
10. Store chemicals safely



CONCLUSION

- Accurate and precise measurement of IAQ is done in PVPSIT.
- Indoor Air Quality Toolkit was used in order to study the Indoor air quality.
- We have evaluated the IAQ parameters(CO_2, Temperature and Humidity) in 167 locations (48 Class Rooms, 68 Staff Rooms & 51 Laboratories) of PVPSIT.
- CO2 showed the widest range of concentrations from 390ppm to 3000 ppm, Temperature from 25-35 degrees Celsius and humidity from 50-75%RH.
- The highest concentrations were measured in the Laboratories of CSE & IT departments due to the lack of ventilation. And at the same time the lowest concentrations were recorded in a staff room of ECE department.
- In parallel with this study, we have conducted a survey on IAQ in PVPSIT, we have received the responses from 400 students all over the campus. This was made our analytical-descriptive study more effective.
- To a lesser extent of our analytical-descriptive study, we will incorporate Internet Of Things (IOT) in our project to monitor the parameters round the clock.

Thank You